

EE 102: Signal Processing and Linear Systems

Problem Set #4: Practice and Study Guide for Midterm Exam #1**Name:** _____**Submission Date:** _____

This problem set is intentionally longer than usual to provide you with more practice problems before the exam. The exam will not be as long as this problem set.

Problem 1 For each question, select the best answer or all correct answers, as directed. Briefly justify your answer.

(a) [3 points] **Select one.** Which expression defines a discrete-time signal?

1. $x(t) = e^{-t}u(t)$
2. $x[n] = \cos(\pi n/4)$
3. $y(t) = x(t - 3)$
4. $\mathcal{H}\{x(t)\} = x(t) + x(t - 2)$

(b) [3 points] **True/False.** The signal $u(t + 2)$ turns on at $t = -2$.

(c) [3 points] **Select all correct answers.** Relative to $x(t)$, the signal

$$y(t) = -4x(3(t + 2))$$

has undergone which transformations?

1. Amplitude scaling by 4
2. Amplitude inversion
3. Time compression by a factor of 3
4. A shift to the left by 2
5. Time reversal

(d) [3 points] **Select one.** What is the energy of $x(t) = 2[u(t) - u(t - 3)]$?

1. 3
2. 4
3. 6
4. 12

(e) [3 points] **True/False.** The system $y(t) = 2x(t) + 1$ is linear.

(f) [3 points] **Select all correct answers.** Which of the following systems are linear and time-invariant?

1. $y(t) = x(t - 2)$
2. $y(t) = t x(t)$
3. $y(t) = |x(t)|$
4. $y(t) = x(t) + x(t - 3)$

(g) [3 points] **Select one.** Evaluate

$$\int_{-\infty}^{\infty} (\tau^2 + 1) \delta(\tau - 3) d\tau.$$

1. 0
2. 3
3. 9
4. 10

(h) [3 points] **True/False.** For $t \geq 0$, the real part of $e^{(-2+j4)t}$ is an oscillating signal whose amplitude decays toward zero.

(i) [6 points] For an LTI system, what is $x(t) * 2\delta(t + 1)$?

[use more pages if needed]

Problem 2 [Adapted from last year's midterm problem] Consider a bacterial cell that divides into two cells every 30 minutes. You start with one cell at $t = 0$. An antibiotic dose is taken every two hours and instantaneously reduces the population to one cell. If a cell division and an antibiotic dose occur at the same time, assume that the antibiotic reset takes priority. Use hours as the time unit.

- (a) [5 points] Describe the bacterial population $x(t)$ as a function of time for $t \geq 0$.
- (b) [5 points] Sketch $x(t)$ over $0 \leq t \leq 8$. Clearly label the times at which the population changes and the population levels.
- (c) [5 points] Extend the same idealized growth-and-reset pattern to all $t \in (-\infty, \infty)$. Is the resulting signal periodic? If it is, find its fundamental period.
- (d) [5 points] Write the equation of $x(t)$ over one period using only constants and shifted unit step functions. Do not use ramps, exponentials, or sinusoids.
- (e) [5 points] Compute the energy of $x(t)$ over one period using the definition of energy for a continuous-time signal.

[use more pages if needed]

Problem 3 Consider the following three continuous-time systems:

$$\mathcal{H}_1\{x(t)\} = x(t - 2),$$

$$\mathcal{H}_2\{x(t)\} = t x(t),$$

$$\mathcal{H}_3\{x(t)\} = x(t) + 1.$$

- (a) [5 points] Determine which systems are linear. Prove linearity or give a counterexample for each system.
- (b) [5 points] Determine which systems are time-invariant. Prove time invariance or give a counterexample for each system.
- (c) [5 points] For every system above that is LTI, draw a system block diagram and find its impulse response.

[use more pages if needed]

Problem 4 An LTI system has input

$$x(t) = u(t) - u(t - 3)$$

and impulse response

$$h(t) = 2\delta(t) + \delta(t - 2) - \delta(t - 4).$$

- (a) [3 points] Sketch $x(t)$ and $h(t)$. Represent each impulse in $h(t)$ with an arrow whose height is its weight.
- (b) [5 points] Write the convolution integral for $y(t) = x(t) * h(t)$ and use the sifting property to express $y(t)$ in terms of shifted copies of $x(t)$.
- (c) [4 points] Write $y(t)$ explicitly using only shifted unit step functions and constants.
- (d) [3 points] Sketch $y(t)$ and clearly label every time at which its value changes.

[use more pages if needed]

Problem 5 Recall that a generalized complex exponential can be written as

$$x_c(t) = Ae^{st},$$

where both constants may be complex. Write

$$A = a_1 + ja_2, \quad s = \sigma + j\omega,$$

where a_1 , a_2 , σ , and ω are real numbers.

(a) [5 points] Find constants a_1 , a_2 , σ , and ω to express $Be^{\sigma t} \cos(\omega t + \phi)$ in the form $x_c(t) = Ae^{st}$.

(b) [10 points] Consider the real-valued signal

$$x_1(t) = 6e^{-2t} \cos\left(4t + \frac{\pi}{3}\right) u(t).$$

Write $x_1(t) = \text{Re}\{Ae^{st}\}u(t)$ and find the corresponding values of a_1 , a_2 , σ , and ω .

[use more pages if needed]